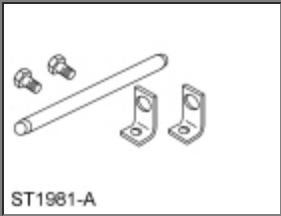
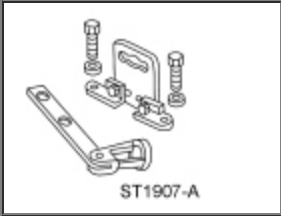
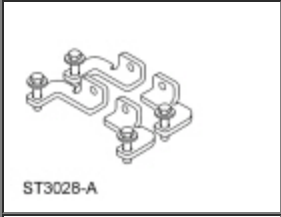


Cylinder Head

 [Ford Ecat Electronic Parts Catalog](#)

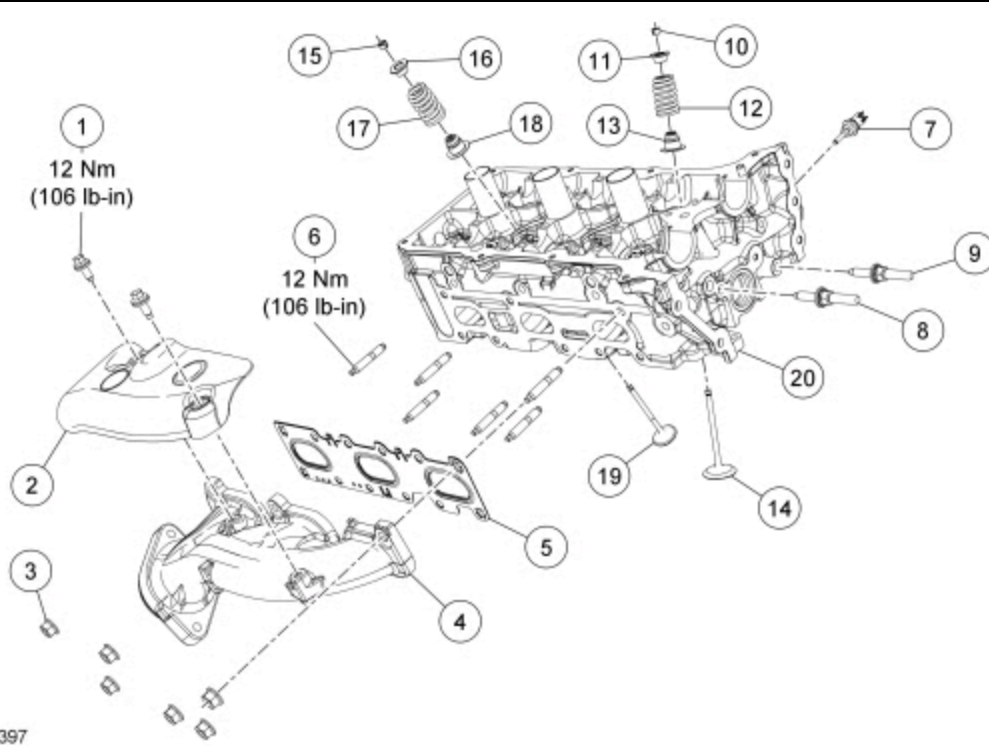
Special Tool(s)

 ST1981-A	Compressor, Valve Spring 303-300 (T87C-6565-A)
 ST1907-A	Compressor, Valve Spring 303-350 (T89P-6565-A)
 ST3028-A	Compressor, Valve Spring 303-1249
 ST1906-A	Installer, Valve Stem Oil Seal 303-470 (T94P-6510-CH)

Material

Item	Specification
Motorcraft® SAE 5W-20 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

NOTE: RH cylinder head shown, LH cylinder head similar.



Item	Part Number	Description
1	W714133 W714133	RH exhaust manifold heat shield bolt (2 required)
2	9A462 9A462	RH exhaust manifold heat shield
3	W701706 W701706	RH exhaust manifold nut (6 required)
4	9430 9430	RH exhaust manifold
5	9448 9448	RH exhaust manifold gasket
6	W712244 W712244	RH exhaust manifold stud (6 required)
7	6G004 6G004	Cylinder Head Temperature (CHT) sensor
8	W714090 W714090	Timing chain tensioner arm pin
9	W714090 W714090	Timing chain guide pin
10	6518 6518	Intake valve spring retainer key (12 required)
11	6514 6514	Intake valve spring retainer (6 required)
12	6513 6513	Intake valve spring (6 required)
13	6571 6571	Intake valve stem seal (6 required)
14	6505 6505	Intake valve (6 required)
15	6518 6518	Exhaust valve spring retainer key (12 required)
16	6514 6514	Exhaust valve spring retainer (6 required)
17	6513 6513	Exhaust valve spring (6 required)
18	6571 6571	Exhaust valve stem seal (6 required)
19	6507 6507	Exhaust valve (6 required)
20	6049 6049	RH cylinder head

Disassembly

All cylinder heads

1. Remove the 2 bolts and the exhaust manifold heat shield.

2. Remove the 6 nuts and the exhaust manifold.

- Discard the nuts and gasket.

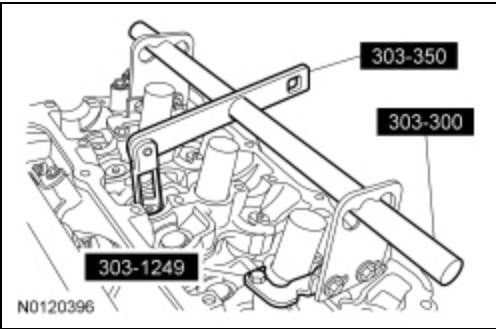
3. Clean and inspect the exhaust manifold. For additional information, refer to Section 303-00.

4. Remove and discard the 6 exhaust manifold studs.

5. Remove the timing chain guide pin.

6. **NOTE:** *If the components are to be reinstalled, they must be installed in the same positions. Mark the components for installation into their original locations.*

Using the Valve Spring Compressors, remove the keys, retainer and spring.



7. Remove the valve from the cylinder head.

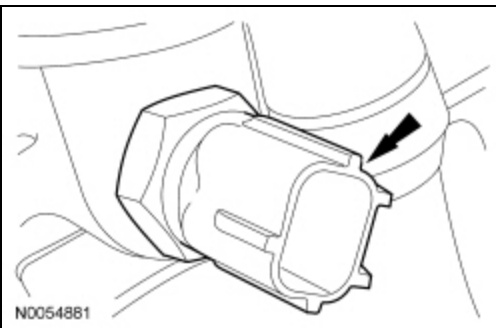
8. Remove and discard the valve stem seal.

9. Repeat the above steps for each valve.

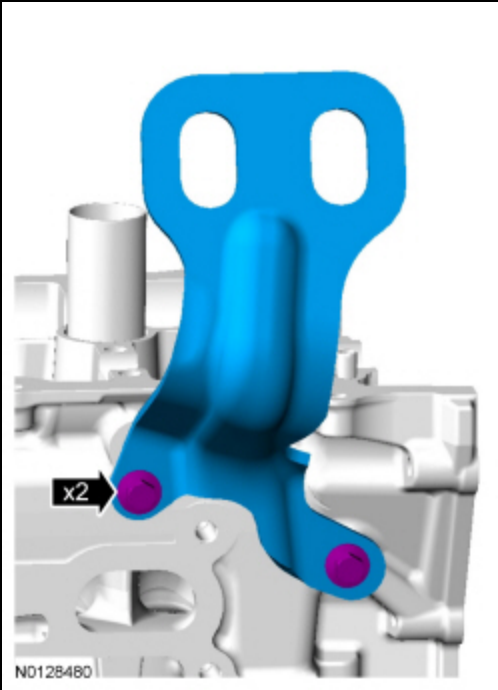
RH cylinder head

10. Remove the timing chain tensioner arm pin.

11. Remove and discard the Cylinder Head Temperature (CHT) sensor.



12. Remove the 2 bolts and the engine lift eye.



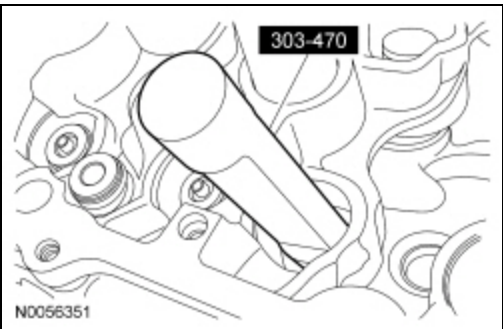
Assembly

All cylinder heads

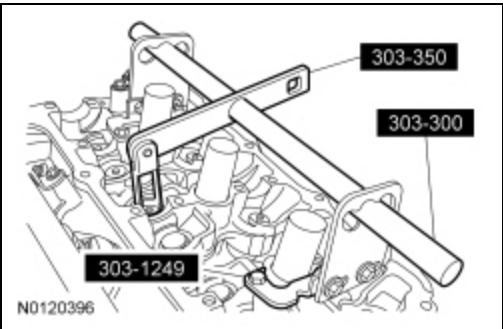
1. **NOTE:** *If the components are to be reinstalled, they must be installed into their original location.*

NOTE: *Lubricate the valve stem and valve stem seal with clean engine oil prior to installation.*

Using the Valve Stem Oil Seal Installer, install a new valve stem seal.



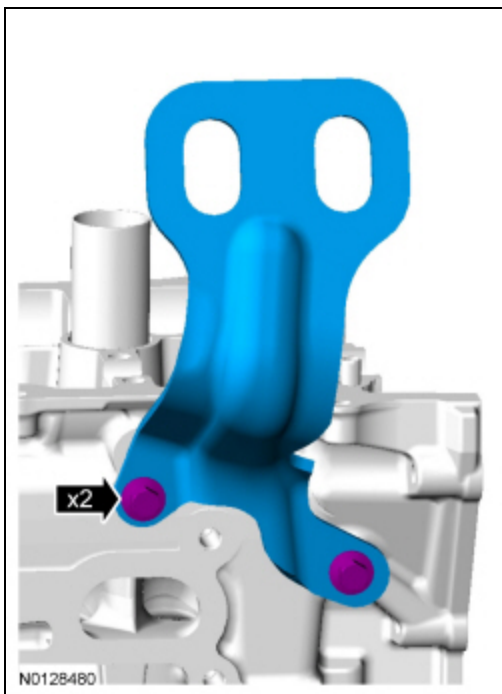
2. Install the valve.
3. Using the Valve Spring Compressors, install the valve spring, retainer and key.



4. Repeat the above steps for each valve.
5. Install the timing chain guide pin and tighten in 2 stages.
 - Stage 1: Tighten to 20 Nm (177 lb-in).
 - Stage 2: Tighten an additional 60 degrees.

RH cylinder head

6. Install the engine lift eye and the 2 bolts.
 - Tighten to 25 Nm (18 lb-ft).

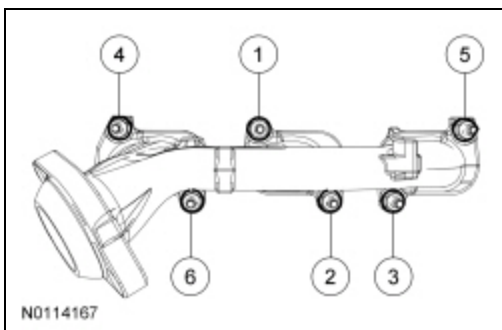


7. Install a new CHT sensor.
 - Tighten to 10 Nm (89 lb-in).
8. Install the timing chain tensioner arm pin and tighten in 2 stages.
 - Stage 1: Tighten to 20 Nm (177 lb-in).
 - Stage 2: Tighten an additional 60 degrees.
9. Install 6 new RH exhaust manifold studs.
 - Tighten to 12 Nm (106 lb-in).

10. **NOTICE: Failure to tighten the exhaust manifold nuts to specification a second time will cause the exhaust manifold to develop an exhaust leak.**

Using a new gasket, install the RH exhaust manifold and 6 new nuts. Tighten in 2 stages in the sequence shown:

- Stage 1: Tighten to 19 Nm (168 lb-in).
- Stage 2: Tighten to 25 Nm (18 lb-ft).



11. Install the RH exhaust manifold heat shield and the 2 bolts.
 - Tighten to 12 Nm (106 lb-in).

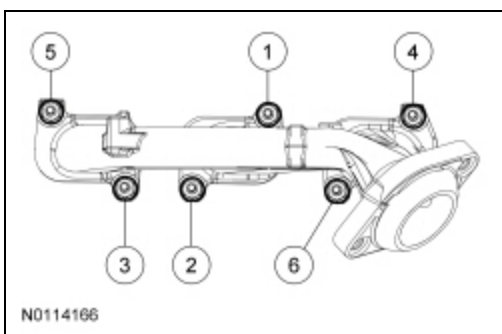
LH cylinder head

12. Install 6 new LH exhaust manifold studs.
 - Tighten to 12 Nm (106 lb-in).

13. **NOTICE:** Failure to tighten the exhaust manifold nuts to specification a second time will cause the exhaust manifold to develop an exhaust leak.

Using a new gasket, install the LH exhaust manifold and 6 new nuts. Tighten in 2 stages in the sequence shown:

- Stage 1: Tighten to 19 Nm (168 lb-in).
- Stage 2: Tighten to 25 Nm (18 lb-ft).



14. Install the LH exhaust manifold heat shield and the 2 bolts.

- Tighten to 12 Nm (106 lb-in).